Statistical Signal Processing

Laboratory 1:
Expectation-Maximization (EM) algorithm

1 Model and data generation

Consider an independent and identically distributed (iid) random sample $X_1, \ldots, X_n$, where the common probability density function (pdf) is

$$X_i \sim p(X_i; \boldsymbol{\theta}) := \epsilon \mathcal{N}(X_i; \mu_1, \sigma^2) + (1-\epsilon)\mathcal{N}(X_i; \mu_2, \sigma^2), \tag{1}$$

and the 3×1 parameter vector is $\boldsymbol{\theta} = [\mu_1, \mu_2, \epsilon]^\top$. Notation $\mathcal{N}(X_i; \mu_j, \sigma^2)$ indicates the Gaussian pdf, namely,

$$\mathcal{N}(X_i; \mu_j, \sigma^2) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{1}{2\sigma^2} (X_i - \mu_j)^2 \right\}.$$

For model (1) we assume that the variance σ^2 is known, hence the only unknown parameters are the weight ϵ and the locations μ_1 and μ_2.

To generate a sample from this model, we proceed as follows: let ϵ_*, $\mu_{1,*}$ and $\mu_{2,*}$ denote the *true* model parameters. Then, in order to draw a realization of X_i, we

1. draw a uniform random variable, $U_i \sim \mathcal{U}(0,1)$, using `u = rand`;
2. if $U_i < \epsilon_*$, then generate $X_i \sim \mathcal{N}(X_i; \mu_1, \sigma^2)$, otherwise generate $X_i \sim \mathcal{N}(X_i; \mu_2, \sigma^2)$. In Matlab, we generate a normal random variable with mean `m` and variance `v` using `x = m + sqrt(v) * randn`.

In order to draw n variables from (1), we

1. generate n uniform variables,

$$\texttt{U} = \texttt{rand}(\texttt{n}, 1);$$

2. compare with the weight ϵ_* (let us denote it as `eps` in Matlab),

$$\texttt{idx1} = \texttt{find}(\texttt{U} < \texttt{eps});$$
$$\texttt{idx2} = \texttt{find}(\texttt{U} > \texttt{eps});$$

3. and sample according to the indices: if the means $\mu_{1,*}$ and $\mu_{2,*}$ are `m(1)` and `m(2)`, respectively, and the variance σ^2 is `v` then

$$\begin{aligned}\texttt{X} &= \texttt{zeros(n, 1)};\\ \texttt{X(idx1)} &= \texttt{m(1)} + \texttt{sqrt(v)} * \texttt{randn(length(idx1), 1)};\\ \texttt{X(idx2)} &= \texttt{m(2)} + \texttt{sqrt(v)} * \texttt{randn(length(idx2), 1)};\end{aligned}$$

2 EM algorithm

The random sample $X_1, \ldots, X_n$ yields the incomplete data. The latent data can be modeled as an iid sample $Y_1, \ldots, Y_n$, where each $Y_i \in \{1, 2\}$ and

$$\begin{aligned}\mathbb{P}(Y_i = 1) &= \epsilon,\\ \mathbb{P}(Y_i = 2) &= 1 - \epsilon.\end{aligned}$$

Then, we can write

$$p(X_i|Y_i; \boldsymbol{\theta}) = \mathcal{N}(X_i; \mu_{Y_i}, \sigma^2),$$

and

$$\begin{aligned}p(X_i; \boldsymbol{\theta}) &= \sum_{j=1}^{2} p(X_i, Y_i = j; \boldsymbol{\theta})\\ &= \sum_{j=1}^{2} p(X_i|Y_i = j; \boldsymbol{\theta})\mathbb{P}(Y_i = j)\\ &= \sum_{j=1}^{2} \mathcal{N}(X_i; \mu_j, \sigma^2)\mathbb{P}(Y_i = j)\\ &= \mathcal{N}(X_i; \mu_1, \sigma^2)\overbrace{\mathbb{P}(Y_i = 1)}^{=\epsilon} + \mathcal{N}(X_i; \mu_2, \sigma^2)\overbrace{\mathbb{P}(Y_i = 2)}^{=1-\epsilon}.\end{aligned}$$

Recall that σ^2 is known; you should set its value for the simulation. If you look at the Matlab code in Section 1, the indices `idx1` and `idx2` determine the realizations of Y_i, $i = 1, \ldots, n$.

Now, let $\mathbf{Y}^n = [Y_1, \ldots, Y_n]^\top$ and $\mathbf{X}^n = [X_1, \ldots, X_n]^\top$. The EM algorithm can be summarized as follows:

i) Initialization: choose $\hat{\boldsymbol{\theta}}_0 = [\hat{\epsilon}_0, \hat{\mu}_{1,0}, \hat{\mu}_{2,0}]^\top$.

ii) Iteration:

- E-step: compute

$$U(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}_k) = \mathbb{E}_{\mathbf{Y}^n|\mathbf{X}^n;\hat{\boldsymbol{\theta}}_k}\left[\log p(\mathbf{X}^n, \mathbf{Y}^n; \boldsymbol{\theta})\right].$$

- M-step: compute

$$\hat{\boldsymbol{\theta}}_{k+1} = \arg\max_{\theta} U(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}_k).$$

Because the samples $\mathbf{X}^n$ and $\mathbf{Y}^n$ are iid, it follows that

$$\begin{aligned} U(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}_k) &= \mathbb{E}_{\mathbf{Y}^n|\mathbf{X}^n;\hat{\boldsymbol{\theta}}_k}\left[\log p(\mathbf{X}^n, \mathbf{Y}^n; \boldsymbol{\theta})\right] \\ &= \mathbb{E}_{\mathbf{Y}^n|\mathbf{X}^n;\hat{\boldsymbol{\theta}}_k}\left[\log \prod_{i=1}^{n} p(X_i, Y_i; \boldsymbol{\theta})\right] \\ &= \sum_{i=1}^{n} \mathbb{E}_{Y_i|X_i;\hat{\boldsymbol{\theta}}_k}\left[\log p(X_i, Y_i; \boldsymbol{\theta})\right]. \end{aligned} \tag{2}$$

Moreover, for any function $h(Y_i)$, we have

$$\mathbb{E}_{Y_i|X_i;\hat{\boldsymbol{\theta}}_k}\left[h(Y_i)\right] = \sum_{j=1}^{2} h(j)\mathbb{P}(Y_i = j|X_i; \hat{\boldsymbol{\theta}}_k).$$

3 Exercises

a) Assume the true parameters $\epsilon_* = \frac{1}{2}$, $\mu_{1,*} = -3$, $\mu_{2,*} = 4$, and $\sigma^2 = 1$. Draw an iid sample of size $n = 100$, then calculate the log-likelihood $\ell_n(\boldsymbol{\theta})$ and plot it for fixed $\epsilon = \epsilon_* = \frac{1}{2}$. Let μ_1 and μ_2 take values between -7 and 7. Use the Matlab functions `mesh` or `contour` for the plot.

Additionally, you must prove that the log-likelihood can be written as

$$\ell_n(\boldsymbol{\theta}) = C + \sum_{i=1}^{n} \log\left[\epsilon \exp\left\{-\frac{1}{2\sigma^2}(X_i - \mu_1)^2\right\} + \right.$$
$$\left. +(1-\epsilon)\exp\left\{-\frac{1}{2\sigma^2}(X_i - \mu_2)^2\right\}\right],$$

where C is constant with respect to (wrt) $\boldsymbol{\theta}$ and, therefore, it can be dropped.

b) Consider the EM method in Section 2. You have to:

1) Prove that

$$U(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}_k) = C + \sum_{i=1}^{n}\sum_{j=1}^{2}\left(\log\left(\mathbb{P}(Y_i = j)\right) - \frac{1}{2\sigma^2}(X_i - \mu_j)^2\right) q_i(j,k),$$

where C is a constant and

$$q_i(j,k) := \frac{\mathcal{N}(X_i; \hat{\mu}_{j,k}, \sigma^2)\hat{\mathbb{P}}_k(Y_i = j)}{\hat{\epsilon}_k \mathcal{N}(X_i; \hat{\mu}_{1,k}, \sigma^2) + (1 - \hat{\epsilon}_k)\mathcal{N}(X_i; \hat{\mu}_{2,k}, \sigma^2)},$$

where $\hat{\mathbb{P}}_k(Y_i = 1) = \hat{\epsilon}_k$ and $\hat{\mathbb{P}}_k(Y_i = 2) = 1 - \hat{\epsilon}_k$.

2) Prove that, at the M-step,

$$\hat{\epsilon}_{k+1} = \frac{1}{n}\sum_{i=1}^{n} q_i(1,k) \quad \text{and} \quad \hat{\mu}_{j,k+1} = \frac{\sum_{i=1}^{n} X_i q_i(j,k)}{\sum_{i=1}^{n} q_i(j,k)}.$$

Hint: Solve the equations $\frac{\partial U(\boldsymbol{\theta},\hat{\boldsymbol{\theta}}_k)}{\partial \epsilon} = 0$ and $\frac{\partial U(\boldsymbol{\theta},\hat{\boldsymbol{\theta}}_k)}{\partial \mu_j} = 0$, $j = 1, 2$.

c) Generate $n = 100$ data points, $X_1, \ldots, X_n$, with (true) parameters $\epsilon_* = \frac{2}{3}$, $\mu_{1,*} = -3$, $\mu_{2,*} = 4$ and $\sigma^2 = 16$. We denote $\boldsymbol{\theta}_* = [\epsilon_*, \mu_{1,*}, \mu_{2,*}]^\top$. Run the EM algorithm with initial values $\hat{\epsilon}_0 = \frac{1}{2}$, $\hat{\mu}_{1,0} = -6$ and $\hat{\mu}_{2,0} = 6$ (and denote $\hat{\boldsymbol{\theta}}_0 = [\hat{\epsilon}_0, \hat{\mu}_{1,0}, \hat{\mu}_{2,0}]^\top$).

Plot (using the commands `semilogy` if/when you need to display very small values)

- the log-likelihood $\ell_n(\hat{\boldsymbol{\theta}}_k)$ versus k,
- the error $|\hat{\epsilon}_k - \epsilon_*|^2$ versus k,
- the errors $|\hat{\mu}_{j,k} - \mu_{j,*}|^2$ versus k, for $j = 1$ and $j = 2$.

1) How many iterations are needed for the algorithm to converge? How do you estimate that number from the simulations?
2) Compare the graph of the log-likelihood and the graphs of the quadratic errors. What do you observe?

d) Run the EM algorithm with the same true parameters $\boldsymbol{\theta}_*$, $\sigma^2 = 16$ and the same initial estimates $\hat{\boldsymbol{\theta}}_0$ but taking $n = 10, 10^2, 10^3, 10^4$. Note that you need to generate new iid data points $X_1, \ldots, X_n$ for each simulation run.

Plot (using the command `semilogy` when needed), on the same axes,

- ϵ_*, $\mu_{1,*}$, $\mu_{2,*}$, $\hat{\epsilon}_k$, $\hat{\mu}_{1,k}$ and $\hat{\mu}_{2,k}$ versus k, for $n = 10, 10^2, 10^3, 10^4$;
- $|\hat{\epsilon}_k - \epsilon_*|^2$ versus k for $n = 10, 10^2, 10^3, 10^4$;
- $|\hat{\mu}_{1,k} - \mu_{1,*}|^2$ versus k, for $n = 10, 10^2, 10^3, 10^4$;
- $|\hat{\mu}_{2,k} - \mu_{2,*}|^2$ versus k, for $n = 10, 10^2, 10^3, 10^4$.

Also plot $\ell_n(\hat{\boldsymbol{\theta}}_k)$ versus k for $n = 10, 10^2, 10^3, 10^4$.

e) With the same setup as in exercise d), estimate and plot the mean square error (MSE)

$$\mathrm{MSE}(n) = \mathbb{E}[\|\boldsymbol{\theta}^n_{MLE} - \boldsymbol{\theta}_*\|^2]$$

versus $n = 10, 10^2, 10^3, 10^4$.

Hint: You will need to generate many (say m) independent estimates $\boldsymbol{\theta}^{n,1}_{MLE}, \boldsymbol{\theta}^{n,2}_{MLE}, \ldots, \boldsymbol{\theta}^{n,m}_{MLE}$ for each value of n and then average the results. Hence, your approximation is $\mathrm{MSE}(n) \approx \frac{1}{m}\sum_{i=1}^{m} \|\boldsymbol{\theta}^{n,i}_{MLE} - \boldsymbol{\theta}_*\|^2$.

f) Repeat exercise d) with the same true parameters $\boldsymbol{\theta}_*$ and $\sigma^2 = 16$, but different initial estimates, $\hat{\epsilon}_0 = 0.5$, $\hat{\mu}_{1,0} = 6$ and $\hat{\mu}_{2,0} = -6$.

Plot (using the command `semilogy` when needed), on the same axes,

- ϵ_*, $\mu_{1,*}$, $\mu_{2,*}$, $\hat{\epsilon}_k$, $\hat{\mu}_{1,k}$ and $\hat{\mu}_{2,k}$ versus k, for $n = 10, 10^2, 10^3, 10^4$;
- $|\hat{\epsilon}_k - \epsilon_*|^2$ versus k for $n = 10, 10^2, 10^3, 10^4$;
- $|\hat{\mu}_{1,k} - \mu_{1,*}|^2$ versus k, for $n = 10, 10^2, 10^3, 10^4$;
- $|\hat{\mu}_{2,k} - \mu_{2,*}|^2$ versus k, for $n = 10, 10^2, 10^3, 10^4$.

Also plot $\ell_n(\hat{\boldsymbol{\theta}}_k)$ versus k for $n = 10, 10^2, 10^3, 10^4$.

Compare with the graphs obtained in question d) and discuss the results.

4 Submission

Prepare a PDF document containing the plots requested in the different exercises and your answers to the questions. For exercise b) you may submit a separate (scanned) handwritten document, also in PDF format. You also need to submit the scripts that you have coded to generate the plots in all the exercises.

Put all documents and scripts in a single compressed file (.zip) and upload it using the link in Aula Global.

You can work on this lab in groups of two students. Nevertheless, both students must submit the (same) files to Aula Global and do not forget to include both names in all the files you submit.